Jinyu Xie

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Education

Sep. 2025 – Present — **Ph.D. in Artificial Intelligence**
Oregon State University, Corvallis, OR.
Advisor: Prof. Xiao Fu.

Sep. 2021 – Jun. 2025 — **B.S. in Mathematics-Physics Fundamental Science**
University of Electronic Science and Technology of China (UESTC), Chengdu, China.
Yingcai Honors College.

Research Experience

Sep. 2025 – Present — **Sparse Autoencoders for Mechanistic Interpretability** — Oregon State University
Advisor: Prof. Xiao Fu.
- Developing DUSAE, a dictionary-learning-inspired sparse autoencoder with decoder-tied proximal-gradient unrolling and feed-forward inference.

Jul. 2022 – Jun. 2025 — **Tensor Factorization and Data Recovery** — UESTC
Advisor: Prof. Xile Zhao.
- Designed tensor-factorization methods for irregular multidimensional data, with applications to spatial transcriptomics and hyperspectral imaging.

Selected Publications

1. **J.-Y. Xie**, H. Zhang, X.-L. Zhao, and Y.-S. Luo, "IRTF: A new tensor factorization for irregular multidimensional data recovery," *Knowledge-Based Systems*, 2025. DOI: 10.1016/j.knosys.2025.114372.
2. Y.-Y. Liu, X.-L. Zhao, **J.-Y. Xie**, Z. Xu, and G. Vivone, "Bi-level tensor decomposition for hyperspectral image restoration," in *IEEE International Geoscience and Remote Sensing Symposium*, 2024. DOI: 10.1109/IGARSS53475.2024.10640494.
3. H. Zhang, T.-Z. Huang, X.-L. Zhao, S.-Q. Zhang, **J.-Y. Xie**, T.-X. Jiang, and M. K. Ng, "Learnable transform-assisted tensor decomposition for spatio-irregular multidimensional data recovery," *ACM Transactions on Knowledge Discovery from Data*, 2024. DOI: 10.1145/3701235.
4. J.-Y. Li, **J.-Y. Xie**, Y.-S. Luo, X.-L. Zhao, and J.-L. Wang, "H2TF for hyperspectral image denoising: Where hierarchical nonlinear transform meets hierarchical matrix factorization," *IEEE Geoscience and Remote Sensing Letters*, vol. 20, 2023. DOI: 10.1109/LGRS.2023.3294933.

Skills

Programming & tools — Python, PyTorch, MATLAB, LaTeX, Git.

Patent

Jul. 2024 — **Co-inventor**, "Irregular high-dimensional data restoration method and system based on transform-domain tensor decomposition," Chinese patent CN117911802B, granted July 9, 2024.

Selected Honors

Jun. 2024 — **Exemplary Student Scholarship**, UESTC (three-time recipient).

Dec. 2023 — **Excellence Award and continued project funding**
University Student Innovation and Entrepreneurship Competition.

May 2023 — **First Prize**, 23rd Mathematical Modeling Contest, UESTC.

Mar. 2023 — **Second Prize**, 14th Chinese Mathematics Competitions for College Students (Mathematics Class A).